$$\mathcal{C}(\tau) = \sum_i g_i^2 \frac{1}{Z_i} \mathrm{Tr}_{\mathcal{E}_i} \left\{ \mathrm{e}^{-\hat{H}_{\mathcal{E}}^{(i)}/k_{\mathrm{B}}T} \hat{\sigma}_z^{(i)}(\tau) \hat{\sigma}_z^{(i)} \right\}, \tag{5.177}$$